


Recall a polynomial in $\mathbb{Z}[x]$ is called primitive if the gcd of its coefficients is 1.

Proposition 1: Let $f(x) \in \mathbb{Z}[x]$ nonzero.

Suppose $f(x) = g(x)h(x)$ in $\mathbb{Q}[x]$.

for some $g(x), h(x) \in \mathbb{Q}[x]$. Then.

there exist $\bar{g}(x), \bar{h}(x) \in \mathbb{Z}[x]$.

such that $f(x) = \bar{g}(x)\bar{h}(x)$

and $\deg \bar{g}(x) = \deg g(x)$

$\deg \bar{h}(x) = \deg h(x)$.

proof. If $f(x)$ is not primitive.

then $\exists ! m \in \mathbb{Z}_+ \text{ s.t. } f(x) = m p(x)$

for some primitive $p(x) \in \mathbb{Z}[x]$.

So it suffices to assume $f(x)$ is primitive.

Since $g(x), h(x)$ has finitely many

rational coefficients, there exist $s', t' \in \mathbb{Z}_+$

i.e. the least common multiples of the

denominator, such that

$$s' \cdot g(x), \quad t' \cdot h(x) \in \mathbb{Z}[x].$$

Then each polynomial can factor out

a primitive part :.e. $\exists s, t \in \mathbb{Z}_+$

such that $s' \cdot g(x) = s \cdot \bar{g}(x)$

$$t' \cdot h(x) = t \cdot \bar{h}(x)$$

where $\bar{g}(x), \bar{h}(x)$ primitive in $\mathbb{Z}[x]$.

$$\begin{aligned} \text{Now } s' t' f(x) &= s' t' g(x) h(x) \\ &= s \cdot t \bar{g}(x) \bar{h}(x). \end{aligned}$$

where $\bar{g}(x) \bar{h}(x)$ is primitive in $\mathbb{Z}[x]$.

It follows $s' t' = s \cdot t$

$$f(x) = \bar{g}(x) \bar{h}(x).$$

It's clear by construction. $\deg \bar{g}(x) = \deg g(x)$
 $\deg \bar{h}(x) = \deg h(x)$

Corollary 2: If $f(x) \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$, then $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Corollary 3. Let $f(x) \in \mathbb{Z}[x]$ be nonconstant.

Then, $f(x)$ is irreducible in $\mathbb{Z}[x]$

iff $f(x)$ is primitive in $\mathbb{Z}[x]$

and $f(x)$ is irreducible in $\mathbb{Q}[x]$.

proof: ($\Rightarrow$) Suppose $f(x)$ is irreducible in $\mathbb{Z}[x]$. then by Corollary 2.

$f(x)$ is irreducible in $\mathbb{Q}[x]$.

Moreover, $f(x)$ is primitive.

Otherwise $f(x) = m \cdot p(x)$ where $m \in \mathbb{Z}_+^*$

and $p(x)$ primitive in $\mathbb{Z}[x]$ with

$\deg p(x) = \deg f(x) > 0$. So $m \cdot p(x)$

is a nontrivial factorization. i.e. none of

the factors is a unit. and $f(x)$

is not irreducible.

($\Leftarrow$). Suppose $f(x)$ is irreducible in $\mathbb{Q}[x]$

We show $f(x)$ is irreducible in $\mathbb{Z}[x]$.

Let $f(x) = g(x)h(x)$ for $g(x), h(x) \in \mathbb{Z}[x]$

View $g(x), h(x)$ in $\mathbb{Q}[x]$.

Since $f(x)$ is irreducible in $\mathbb{Q}[x]$.

WLOG, we assume $\deg g(x) = 0$

i.e. $g(x) \in \mathbb{Z}_+$ Since $f(x)$ is

primitive. $g(x) = 1$, which implies

$f(x)$ is irreducible in $\mathbb{Z}[x]$.

Remark: NOT all irreducible polynomial

in $\mathbb{Q}[x]$ is irreducible in $\mathbb{Z}[x]$.

eg. $3x-3 = 3(x-1)$. But the only

thing that could go wrong is the

nonprimitivity. However, it is not

obvious that irreducible in $\mathbb{Z}[x]$ remains irreducible in $\mathbb{Q}[x]$ since $\mathbb{Z}[x]$ is a smaller ring and seemingly contains less possibility for factorization.

Corollary 4: All irreducible polynomials in $\mathbb{Z}[x]$ are.

(1) prime numbers in $\mathbb{Z}$

(2) primitive polynomials that

is irreducible in $\mathbb{Q}[x]$.

Corollary 5: $\mathbb{Z}[x]$ is UFD.

proof: WLOG let $f(x) \in \mathbb{Z}[x]$ be primitive.

We first show $f(x)$ admits an irreducible factorization of finite length.

View $f(x) \in \mathbb{Q}[x]$ where $\mathbb{Q}[x]$ is a UFD.

Then $f(x) = f_1(x) \cdots f_n(x)$ for

$f_i(x)$ irreducible in $\mathbb{Q}[x]$

For each $f_i(x)$, there exists $\lambda_i' \in \mathbb{Z}_+$

s.t. $\lambda_i' f_i(x) \in \mathbb{Z}[x]$ and

write $\lambda_i' f_i(x) = \lambda_i \overline{f_i(x)}$

for $\lambda_i \in \mathbb{Z}_+$ and $\bar{f}_i(x)$ primitive

in $\mathbb{Z}[x]$. Then $(\prod_{i=1}^n \lambda_i) f(x) = \prod_{i=1}^n \lambda_i \bar{f}_i(x)$

in $\mathbb{Z}[x]$. Since $f(x)$ is primitive

and $\prod_{i=1}^n \bar{f}_i(x)$ primitive, we conclude

$$f(x) = \bar{f}_1(x) \cdots \bar{f}_n(x).$$

Now we claim $\bar{f}_i(x)$ is irreducible

in $\mathbb{Z}[x]$. But each $\bar{f}_i(x)$ is primitive

and irreducible in $\mathbb{Q}[x]$. So by Corollary 3

$\bar{f}_i(x)$ is irreducible in $\mathbb{Z}[x]$.

This shows the existence.

It remains to show the uniqueness.

$$\begin{aligned}\text{Suppose } f(x) &= f_1(x) \cdots f_n(x) \\ &= g_1(x) \cdots g_m(x)\end{aligned}$$

are two irreducible factorization in $\mathbb{Z}[x]$.

Since we assumed $f(x)$ is primitive

we may assume all of $f_i(x), g_j(x)$

are nonconstant polynomials and

are primitive. Then $f_i(x), g_j(x)$

are irreducible in $\mathbb{Q}[x]$. The uniqueness

follows from $\mathbb{Q}[x]$ being a UFD